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Cultured *Cordyceps sinensis* polysaccharides modulate intestinal mucosal immunity and gut microbiota in cyclophosphamide-treated mice

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ABSTRACT

The present study aimed to investigate the protective effect of cultured *Cordyceps sinensis* polysaccharides (CSP) on cyclophosphamide (Cy)-induced intestinal mucosal immunosuppression and microbial dysbiosis in mice. Results showed that CSP stimulated cytokines secretion (IL-12, IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, TGF-β3, TNF-α, IL-2, IL-21) and transcription factors production (T-bet, GATA-3, RORγt, Foxp3). TLRs (TLR-2, TLR-4, TLR-6) and NF-κB pathway key proteins (p-IκB-α, NF-κB p65) were also upregulated after CSP administration. Moreover, CSP recovered SCFAs levels which decreased by Cy treatment. Furthermore, 16S rRNA sequencing of fecal samples was performed. α-diversity and β-diversity analysis revealed CSP improved microbial community diversity and modulated the overall structure of gut microbiota. Taxonomic composition analysis found that CSP increased the abundance of probiotics (*Lactobacillus, Bifidobacterium, Bacteroides*) and decreased pathogenic bacteria (*Clostridium, Flexispira*). These findings suggested the potential of CSP as a prebiotics to reduce side effects of Cy on intestinal mucosal immunity and gut microbiota.

1. Introduction

Cyclophosphamide (Cy) is one of the famous anticancer agent which remains extensively used in the treatment of haematological malignancies and various epithelial tumours (Ahlmann & Hempel, 2016). However, Cy has diverse side effects, resembling that of other chemotherapeutic agents. Cy at high doses will damage the gastrointestinal mucosal, leading to the increased risk of immunodeficiency and secondary infections (Yu et al., 2015). Moreover, it was demonstrated that Cy could alter gut microbiota and induce the translocation of specific Gram-positive bacteria into secondary lymphoid organs (Viaud, Saccheri, Mignot, Yamazaki, & Daillère, 2011). Therefore, in order to reduce the side effects of Cy and improve the curative effect, great efforts have been made to develop immunomodulating agents.

Intestine is the largest digestive and absorbing organ, as well as an important defensive barrier in human body. While exerting the absorptive functions, mucosal barriers of a healthy body should also be able to protect the host against toxins with pathogenic capacities and invasion of microbes (Bischoff & Kra¨mer, 2007). A great deal of previous researches have proved that $CD4^+$ T cells are vital for host defense and intestinal mucosal immunity, which can orchestrate the full panoply of immune responses by producing cytokines and chemokines (Zhu & Paul, 2008). There are four main T helper (Th) cells: Th1, Th2, Th17 and regulatory T (Treg) cells. Th1 cells produce interferon-gamma (IFN-γ) and tumor necrosis factor alpha (TNF-α) as signature cytokines and be identified as critical for cell mediated inflammation (Yu et al., 2014). Th2 cells generate interleukin (IL)-4, IL-5, IL-13 and protect the host from the invasion of extracellular pathogens. Most of they function in epithelial tissues, especially in lungs and intestinal tract (Licona-Limon, Kim, Palm, & Flavell, 2013). Th17 cells are characterized based on the secretion of IL-17A, IL-17 F and IL-22, which participate in host defense against fungi, extracellular bacteria and other eukaryotic

Abbreviations: CSP, cultured *Cordyceps sinensis* polysaccharides; Cy, cyclophosphamide; Foxp3, forkhead box P3; Gata-3, GATA-binding transcription factor 3; IFN, interferon; LH, levamisole hydrochloride; IL, interleukin; LDA, linear discriminant analysis; LEfSe, linear discriminant analysis effect size; NCSP, natural *Cordyceps sinensis* polysaccharides; NMDS, nonmetric multidimensional scaling; OTUs, operational taxonomic units; PCA, principal component analysis; RORγt, RAR-related orphan receptor gamma t; SCFA, short-chain fatty acid; T-bet, T box transcriptional factor; TGF, transforming growth factor; TLR, toll-like receptor; TNF, tumor necrosis factor

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pathogens (Korn, Bettelli, Oukka, & Kuchroo, 2009). Furthermore, IL-10 and transforming growth factor-beta (TGF-β) are mostly associated with Treg cells suppressive functions, Treg cells are regarded as important immunoregulators in many autoimmune disease conditions and inflammatory (Xie et al., 2016). T-bet, GATA-3, RORγt and Foxp3 are the master transcription factors of Th1, Th2, Th17 and Treg cells differentiation, respectively (Raphael, Nalawade, Eagar, & Forsthuber, 2015).

Gut microbiota is intimately related to intestinal mucosal immunity that microbiota may exert both pro- and anti-inflammatory responses. Gut microbiota inhabiting the human intestinal tract constitute a complex ecosystem, which contain about 10^{14} bacteria of over 1000 different species (Rajilic-Stojanovic & de Vos, 2014). Besides significantly contributing to the digestion, gut microbiota can also produce short-chain fatty acids (SCFAs) which is of great importance to regulate timely immune responses and eliminate inflammation (Cryan & Dinan, 2012; Rajilic-Stojanovic, 2013). A diverse composition of gut microbiota is important for the host, which is conducive to the self-regulation of intestinal microbial ecology. In adults, *Bacteroidetes* and *Firmicutes* are generally the main components of gut microbiota, whereas *Proteobacteria*, *Actinobacteria* and *Verrucomicrobia* are usually minor constituents although found in many people. Most intestinal microbes are either harmless or beneficial to the host (Lozupone, Stombaugh, Gordon, Jansson, & Knight, 2012).

Polysaccharides are complex carbohydrates with various bioactivities (Chen et al., 2014; Chen, Zhang, Gao, Huang, & Wu, 2017). Recent years, numerous studies indicated that polysaccharides had the ability of regulating intestinal mucosal immunity (Duerkop, Vaishnava, & Hooper, 2009). There were also researches showed that non-digestible polysaccharides could be recognized as potential prebiotics for the utilization of gut microbiota, leading to the alteration of microbial diversity and composition (Porter & Martens, 2017). For example, polysaccharides from *Dendrobium huoshanense* could stimulate cytokines production, induce the immune cells proliferation and differentiation in the intestine, and enhance the relative abundance of *Lactobacillus, Parabacteroides, Prevotella, Porphyromonas*, reduce *Helicobacter, Clostridium* in mice colon (Xie et al., 2019).

Cordyceps sinensis (*C. sinensis*), a valuable parasitic fungus, has been widely used as herbal medicine and functional food in China for hundreds of years. Polysaccharides are one of the main bioactive components of *C. sinensis* (Nie, Cui, Xie, Phillips, & Phillips, 2013). Recently, we purified three kinds of polysaccharides from three natural *C. sinensis* (NCSP) at different altitudes in Qinghai province of China and one kind of polysaccharides from cultured *C. sinensis* (CSP). Experiments showed that the composition and structure of three NCSPs were similar while CSP was different with them, and they all had the ability to scavenge DPPH and hydroxyl radicals. Because the productivity of cultured *C. sinensis* was much higher and the price was much lower, we chose CSP as the material for further research. CSP had high purity, which was mainly composed of glucose, galactose, mannose, galacturonic acid, arabinose as well as trace proteins and phenolic compounds. The backbone of CSP was 1,4-glucose and 1,4-galactose, with a molecular weight of 28 kDa approximately (Table S1) (Wang et al., 2017). The latest study demonstrated that CSP could improve intestinal morphology, restore intestinal related enzymes, growth factor and its receptor, regulate tight junctions and so on in Cy-treated mice (Ying et al., 2019). Therefore, based on the protective effects of CSP on intestinal barrier, we hypothesized that CSP might have a positive function on intestinal mucosal immunosuppression and microbial dysbiosis. The present study was further designed to testify the modulation effect of CSP on the intestinal mucosal immunity and gut microbiota in Cy-treated mice.

2. Materials and methods

2.1. Materials and reagents

Cultured *Cordyceps sinensis* was obtained from Guoyao Company Jiangxi, China. Cyclophosphamide No. 53832137 was from Sigma-Aldrich St. Louis, MO, USA. Levamisole hydrochloride (LH) was purchased from Shandong Renhe Church Pharmaceutical Co., Ltd. Tancheng, Shandong, China. Mouse IL-12, IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, TGF-β3, TNF-α, IL-2 and IL-21 ELISA kits were from Boster Biological Technology Co., Ltd. Wuhan, Hubei, China. FastQuant RT Kit With gDNase and SuperReal PreMix Plus SYBR Green were purchased from TIANGEN Biotech Co., Ltd (Beijing, China). Antibodies against T-bet, GATA-3, Foxp3, NF-κB p65, phospho-IκB-α (p-IκB-α) were from Cell Signaling Technology (Beverly, MA, USA). Anti-RORγt antibody was obtained from BD Biosciences (San Jose, CA, USA). Anti-histone H2AX antibody was purchased from abcam (Shanghai, China). The β-actin and related secondary antibodies were purchased from ZSGB Biotechnology (Beijing, China). EZNA™Mag-Bind Soil DNA Kit was from Omega bio-tek (Shanghai, China) AxyPrep DNA Gel Extraction Kit was purchased from Axygen (NY, USA) TruSeq Nano DNA LT Library Prep Kit was obtained from Illumina (SD, USA).

2.2. Animals and experimental design

Animal experiments were approved by the Institutional Animal Care and Use Committee, Nanchang University. Six to eight-week-old specific pathogen free (SPF) female BALB/c mice weighing 20.0 ± 2.0 g with the animal certificate SCXK (Xiang) 2016-0002 were bought from Hunan Slac Jingda Labora tory Animal Co. Ltd. (Changsha, China). The mice were kept at temperature 22 ± 2 ℃, 12 h/12 h light/dark cycle, 55 ± 10 % relative humidity and provided with adlibitum food and water.

After one week acclimatization, all mice were assigned to six groups (n = 8 in each group) randomly: normal control group (NC), Cy model group (M), Cy + low-dose CSP group (LD), Cy + medium-dose CSP group (MD), Cy + high-dose CSP group (HD) and positive control group (PC). All mice except for NC group were subjected to intraperitoneal injection of 80 mg/kg body weight (BW) Cy once a day for three days while mice in NC group were injected with normal saline. During the remaining seven days, mice in NC group and M group were treated with normal saline once a day, and mice in LD group, MD group, HD group were administrated with CSP at 25, 50, 100 mg/kg BW, respectively. Meanwhile, mice in PC group were gavaged with 40 mg/kg BW LH (Fig. S1). Fresh feces were collected on the 10th day, and the mice were sacrificed on Day 11. The small intestinal tissue and the contents of colon were excised for subsequent analysis.

2.3. Polysaccharides preparation

The extraction and purification of CSP were conducted as described previously (Wang et al., 2017). Briefly, CSP were prepared by water extraction and alcohol-precipitation. Cultured *Cordyceps sinensis* powder was first soaked in 80 % (v/v) ethanol and extracted twice with deionized water (1:20, w/v, 95–100 ℃). Then, the crude polysaccharides were deproteinized using Sevag method (Staub, 1965). After dialysis, the polysaccharide solution was freeze-dried and CSP were therefore obtained.

2.4. Cytokine assays

The supernatant of small intestinal tissue (100 mg) was collected after homogenizing over ice in 900 u L of phosphate buffer solution (PBS) and centrifugating (8000 ×*g*, 15 min, 4 ℃). The levels of cytokines IL-12, IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, TGF-β3, TNF-α, IL-2 and IL-21 were measured by ELISA kits based on the instructions

Table 1
Effects of CSP on the levels of cytokines in intestine of mice.

Group	NC	M	LD	MD	HD	PC
IL-12 (pg/mL)	20.49 ± 1.30	4.02 ± 2.94##	29.53 ± 6.59**	14.34 ± 1.91**	16.03 ± 5.63**	14.48 ± 4.53**
IFN-γ (pg/mL)	38.46 ± 6.66	14.92 ± 4.82##	31.2 ± 6.65*	41.28 ± 13.03**	46.78 ± 8.12**	32.02 ± 5.71**
IL-4 (pg/mL)	17.01 ± 1.19	12.33 ± 1.47##	14.21 ± 0.27	15.92 ± 2.22*	17.08 ± 3.89**	15.42 ± 2.26*
IL-13 (pg/mL)	6.35 ± 1.11	2.78 ± 0.31#	9.73 ± 0.73**	11.42 ± 1.58**	30.50 ± 9.60**	8.95 ± 2.79**
IL-6 (pg/mL)	121.44 ± 29.88	25.46 ± 13.07##	31.49 ± 10.5	100.42 ± 22.79**	128.99 ± 36.2**	61.9 ± 25.15*
IL-17 (pg/mL)	79.51 ± 15.98	26.51 ± 5.95##	50.81 ± 15.23*	55.9 ± 14.82**	73.04 ± 10.52**	65.32 ± 7.83**
IL-10 (pg/mL)	158.3 ± 49.04	26.35 ± 8.64##	52.26 ± 12.82*	35.07 ± 15.43	112.26 ± 30.9**	57.87 ± 20.82*
TGF-β3 (pg/mL)	238.26 ± 38.05	95.1 ± 34.4##	149.33 ± 42.89*	220.09 ± 44.11*	219.51 ± 36.88*	141.42 ± 17.92
TNF-α (pg/mL)	84.4 ± 15.22	18.52 ± 4.27##	46.83 ± 16.33*	92.98 ± 22.53**	62.28 ± 18.02**	71.17 ± 18.11**
IL-2 (pg/mL)	29.95 ± 8.76	7.47 ± 2.34##	14.49 ± 3.23*	19.63 ± 3.1**	22.94 ± 4.64**	16.15 ± 3.76**
IL-21 (pg/mL)	210.73 ± 28.69	109.31 ± 27.02##	131.14 ± 23.82	185.27 ± 31.49**	157.48 ± 36.81*	173 ± 27.88**

Data are expressed as the mean ± SD (n = 8). #$p < 0.05$, ##$p < 0.01$ compared with NC group; *$p < 0.05$, **$p < 0.01$ compared with M group.

provided by manufacturer.

2.5. RNA extraction and RT-qPCR

Total RNA was extracted from small intestinal tissue with TRIzol reagent (Beyotime, Shanghai, China) following the manufacturer's instruction, and cDNA was obtained by FastQuant RT Kit (With gDNase) according to the manufacturer's protocol. Then, the qPCR was performed using the QuantStudio™ 7 Flex Real-Time PCR System (Applied Biosystems, CA, USA) with SuperReal PreMix Plus (SYBR Green). Data were analyzed by $2^{-\Delta\Delta Ct}$ method. The β-actin was used as the housekeeping gene for normalization. Primer sequences used were shown in Table S2.

2.6. Western blot analysis

Proteins of cytoplasm or nuclear fractions were obtained from small intestinal tissue by extraction kit (Beyotime, Shanghai, China). Protein concentration was detected using the BCA assay kit (Beyotime, Shanghai, China). Then, 30 μg of denatured proteins were separated on 8 ~ 12 % SDS-PAGE gels. After the transference of proteins, PVDF membrane (Millipore Co., MA, USA) was blocked with 5 % BSA. Membranes were incubated in specific primary antibodies and then incubated with HRP-conjugated secondary antibodies. Proteins were checked by ECL Plus detection kit, and protein densitometric was performed by Quantity One 4.6.2 (Bio-Rad Laboratories, CA, USA).

2.7. Determination of SCFAs in the contents of colon

SCFAs concentrations in colonic contents were detected using an Agilent 6890 N GC system (Agilent, USA) with a flame ionization detector (FID) as described previously (Hu, Nie, Min, & Xie, 2012). Briefly, 100 mg of colonic content for each sample was diluted with deionized water with the ratio of 1 : 6, vortexed for 2 min, ultrasound for 5 min, and then stand in the ice-cold water bath for 20 min. After centrifugation 10,000 ×*g*, 20 min, 4 °C and repeated once, the supernatants were collected and filtered for futher analysis.

2.8. 16S rRNA sequencing of gut microbiota

Total DNA was isolated from mouse fecal samples using EZNA™Mag-Bind Soil DNA Kit (Omega bio-tek, USA) reference to the manufacturer's recommendations. The V3–V4 regions of bacterial 16S rRNA gene was amplified by PCR, with universal primers 338 F (5′-ACTCCTACGGGAGGCAGCA-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′). Subsequently, duplicate PCR products were excised from a 2 % agarose gel and purified by AxyPrep DNA Gel Extraction Kit (Axygen, NY, USA). Sequencing library was build up using TruSeq Nano DNA LT Library Prep Kit (Illumina, SD, USA) and then sequenced by an Illumina MiSeq platform (Illumina, SD, USA).

2.9. Bioinformatics analysis

Paired-end reads from the same original DNA were assembled by FLASH (v1.2.7, http://ccb.jhu.edu/software/FLASH/) and then the high-quality sequences were clustered into operational taxonomic units (OTUs) according to a 97 % threshold of pairwise identity using the uclust function in QIIME (v1.8.0, http://qiime.org/). Greengenes database (Release 13.8, http://greengenes.secondgenome.com/) was further used as a reference database to annotate the sequences. The within-community diversity (α-diversity) was analysed by calculating Simpson, Chao, ACE and Shannon indices. Variation in community composition (β-diversity) was measured by calculating the principal component analysis (PCA), weighted and unweighted uniFrac non-metric multidimensional scaling (NMDS) analysis, and hierarchical clustering analysis based on weighted UniFrac distances. Dominant bacterial relative abundance of at the phylum and genus levels were also analysed. They were all performed by QIIME and R software. The linear discriminant analysis (LDA) and LDA effect size (LEfSe) methods were applied to compare and visualize different taxa microbes among groups based on Galaxy Online Analysis Platform (http://huttenhower.sph.harvard.edu/galaxy/).

2.10. Statistical analysis

All values in the figures and tables are presented as mean ± SD. Data were analyzed using ANOVA followed by *post hoc* LSD test. SPSS statistical software 23 (SPSS Inc., Chicago, IL, USA) was used for performing statistical analysis. A value of $p < 0.05$ is considered as statistically significant.

3. Results

3.1. Effects of CSP on the levels of cytokines in small intestinal tissue

The contents of IL-12, IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, TGF-β3, TNF-α, IL-2 and IL-21 in different groups of mice were shown in Table 1. Compared with NC group, all cytokines levels decreased significantly in M group ($p < 0.05$, $p < 0.01$). Simultaneously, results showed that CSP played positive roles in the recovery of cytokines, especially IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, and IL-2 in HD group. In PC group, LH administration also restored the levels of all cytokines except TGF-β3 ($p < 0.05$, $p < 0.01$).

3.2. CSP improved the mRNA levels of TNF-α, IL-2 and IL-21 in Cy-treated mice

As shown in Fig. 1, M group displayed lower mRNA levels of TNF-α, IL-2 and IL-21 compared with NC group. However, obvious enhancements appeared after CSP treatment at all three different doses except low dose for IL-21 ($p < 0.05$, $p < 0.01$). Moreover, TNF-α, IL-2 and IL-

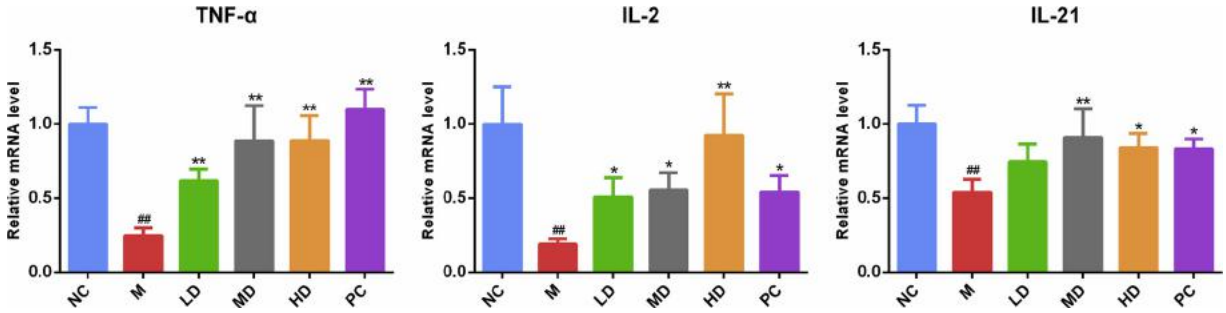

Fig. 1. CSP improved the mRNA levels of TNF-α, IL-2 and IL-21 in Cy-treated mice. Data are expressed as the mean ± SD (n = 8). $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with NC group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with M group.

21 mRNA expression were all restored as well in PC group.

3.3. Effects of CSP on the expression and ratio of T-bet, GATA-3, RORγt, Foxp3

To investigate whether CSP were associated with Th cells, we examined the protein levels of Th1 cell-specific transcription factor T-bet, Th2 cell-specific transcription factor GATA-3, Th17 cell-specific transcription factor RORγt and Treg cell-specific transcription factor Foxp3 by western blotting. Results showed that the levels of T-bet, GATA-3, RORγt and Foxp3 in LD, MD, HD and PC groups were higher than that of M group (Fig. 2A–D).

3.4. CSP regulated TLRs mRNA expression and NF-κB pathway key proteins

As shown in Fig. 3, Cy resulted in a remarkable decline of TLR-2, TLR-4, TLR-6 and TLR-9 mRNA expression ($p < 0.05$, $p < 0.01$). In contrast, the CSP treatment increased the mRNA levels of TLR-2, TLR-4 and TLR-6 ($p < 0.05$, $p < 0.01$). However, the TLR-9 mRNA expression displayed no significant different in three CSP-treated groups as well as PC group. Furthermore, western blotting results showed the cytoplasmic expression of p-IκB-α and the nuclear expression of NF-κB p65. The trend of the two proteins in different groups were similiar which decreased markedly in M group and reversed significantly after CSP treatment ($p < 0.05$, $p < 0.01$). For both p-IκB-α cytoplasmic protein and NF-κB p65 nuclear protein, high-dose CSP administration had the best recovery effect. PC group also upregulated both protein levels, but

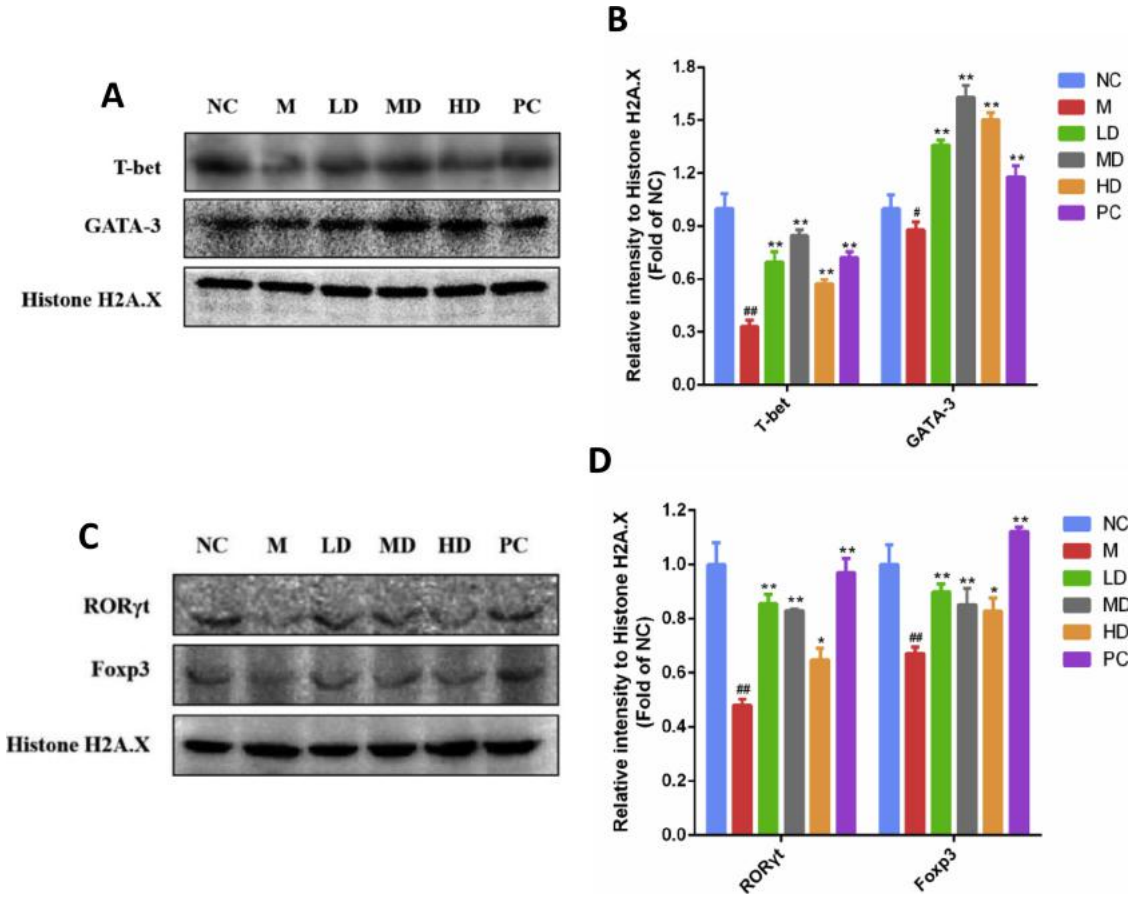

Fig. 2. CSP promoted the expression of T-bet, GATA-3, RORγt, Foxp3 in Cy-treated mice. Proteins of nuclear fractions were extracted. (A) The protein levels of T-bet and GATA-3. (B) Relative intensities of T-bet and GATA-3 bands. (C) The expression of RORγt and Foxp3. (D) Relative intensities of RORγt and Foxp3 bands. Histone H2A.X was used as a nuclear marker. Data are expressed as the mean ± SD (n = 3). $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with NC group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with M group.

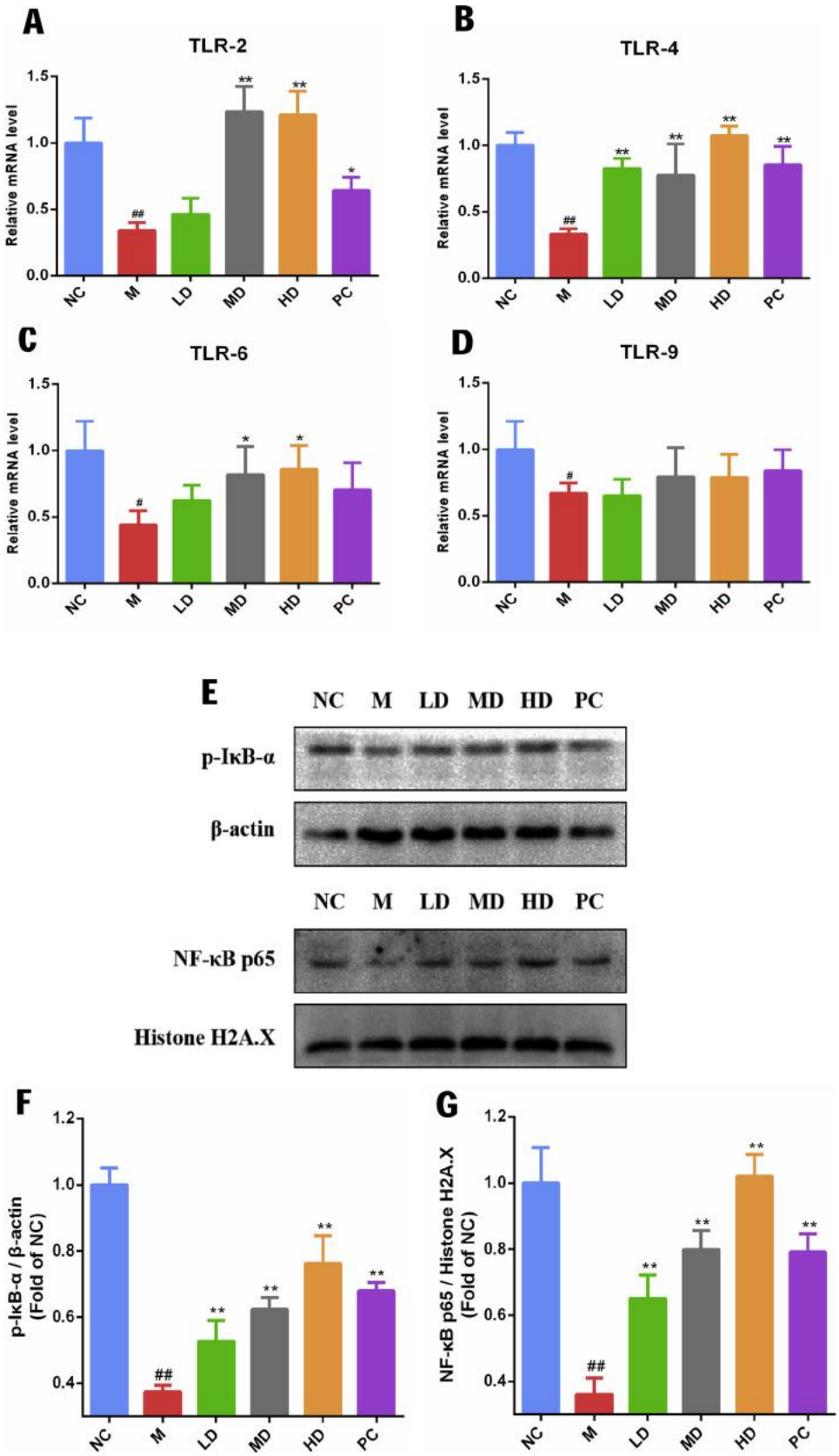

Fig. 3. CSP regulated TLRs mRNA expression and NF-κB pathway key proteins. mRNA was isolated from small intestinal tissue, and proteins of cytoplasm or nuclear fractions were extracted, respectively. (A–D) The mRNA expression of TLR-2, TLR-4, TLR-6 and TLR-9. (E) Western blot bands of p-IκB-α and NF-κB p65. (F–G) Relative intensities of p-IκB-α and NF-κB p65 bands. β-actin was an endogenous control and histone H2A.X was used as a nuclear marker. Data are expressed as the mean ± SD (n = 8 of A–D and n = 3 of E–G). $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with NC group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with M group.

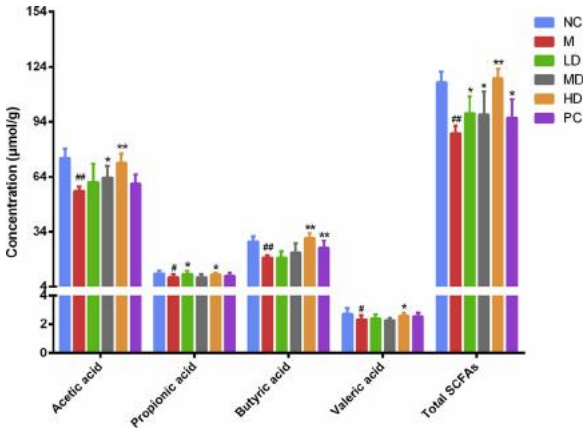

Fig. 4. Effects of CSP on the SCFAs in the contents of colon. Data are expressed as the mean ± SD (n = 8). $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with NC group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with M group.

the effect was not as good as HD group (Fig. 3F–G).

3.5. Effects of CSP on the SCFAs in the contents of colon

To determine the effect of CSP on SCFAs generated by microbiota, the concentrations of acetic acid, propionic acid, butyric acid, valeric acid and total SCFAs in the contents of colon were measured. From Fig. 4, we could find that SCFAs decreased in M group compared with NC group ($p < 0.05$, $p < 0.01$). However, noticeable enhancements could be seen in all kinds of acids and total SCFAs after high-dose CSP treatment. The total SCFAs increased obviously in LD, MD, HD and PC groups compared with M group ($p < 0.05$, $p < 0.01$).

3.6. CSP modulated the overall structure of gut microbiota in Cy-treated mice

As the data of SCFAs indicated that HD group showed better effects in the Cy immunosuppressed animal models, total DNA was extracted from NC group, M group and HD group, which renamed as CSP group in the following experiments, fecal samples for further 16S rRNA sequencing to investigate gut microbiota. We collected 601,495 highly qualified reads in total, 33,416 reads on average for each sample n = 6 per group. According to a 97 % threshold of pairwise identity, OTUs were defined. As shown in Table 2, the number of OTUs in NC group and M group had no significant difference but increased obviously in CSP group in contrast to M group. Microbial community richness was analysed by measuring Chao1 and ACE indices (Su et al., 2018). Microbial community diversity and uniformity were described by detecting Shannon and Simpson indices (Su et al., 2018). The data showed that Chao1, ACE and Shannon indices in CSP group were both much higher than that in M group. Additionally, we investigated the overall microbial community structure similarity by analysing β-diversity, including PCA (Fig. 5A), NMDS analysis (Fig. 5B–C) and Hierarchical clustering analysis (Fig. 5D). The top two principal components in PCA accounted for 55.92 % and 33.78 % of variation in the total data, respectively. PCA result displayed that the microbial community structure in CSP group was significantly different from M group and much closer to NC group. Similar observation could also be obtained in NMDS analysis and Hierarchical clustering analysis.

3.7. Effects of CSP on microbial community composition at the phylum and genus levels

To further understand the specific changes of microbial community, we analyzed the taxonomic composition of each group samples. At the phylum level, microbiota were mainly composed of *Firmicutes*, *Bacteroidetes*, *Proteobacteria* and *Actinobacteria*, among which *Firmicutes* and *Bacteroidetes* accounted for more than 90 % of them (Fig. 6A). As we could see in Table S3, Cy treatment would increase the abundance of *Firmicutes* and *Proteobacteria* and decrease the abundance of *Bacteroidetes*. In contrast, CSP reversed these changes, *Firmicutes*, *Proteobacteria* as well as *Actinobacteria* downregulated and *Bacteroidetes* upregulated. At the genus level, *Lactobacillus*, *Unclassified_S24-7*, *Unclassified_Clostridiales*, *Unclassified_Lachnospiraceae*, *Bacteroides*, *Flexispira*, *Oscillospira*, *Adlercreutzia*, *Bifidobacterium*, *Ruminococcus* and *Clostridium* were the eleven most predominant microbiota (Fig. 6B). Among eight classified genus, the abundance of *Lactobacillus* was the highest. In addition, *Lactobacillus*, *Bacteroides*, *Bifidobacterium* obviously decreased and *Flexispira*, *Clostridium* greatly raised while *Adlercreutzia*, *Oscillospira* showed no difference in M group. However, CSP treatment could weaken the influences of Cy, the abundance of *Lactobacillus*, *Bacteroides*, *Bifidobacterium* recovered and *Flexispira*, *Clostridium* reduced (Table S4).

3.8. Key phylotypes of gut microbiota altered by CSP

To identify the key phylotypes of gut microbiota in different groups, we performed LEfSe analysis among NC group, M group and CSP group. LDA results showed ten discriminative features in M group (LDA > 3.6, $p < 0.05$), and *Bacteroidetes*, *Bacteroidia*, *Bacteroidales* were the main microbiota. CSP group showed ten dominant microorganisms as well (LDA > 3.6, $p < 0.05$), and the major microbiota were *Firmicutes*, *Clostridia*, *Clostridiales*. However, there were no dominant microorganisms in NC group according to the analysis (Fig. 6C). Then, an evolutionary clustering analysis diagram was delivered to identify major microflora by taxonomy (Fig. 6D). In cladogram, *Bacteroidetes* and *Verrucomicrobiae* had the highest flora abundance in the green parts and *Firmicutes* was the richest in the red area, which represented CSP group and M group, respectively. Overall, these results indicated that CSP treatment altered the key phylotypes of gut microbiota in Cy-treated mice faeces and promoted the multiplication of specific bacteria.

4. Discussion

Intestine is the largest digestive and absorbing organ, known as an important mucosal barrier in human body. Cy as an anticancer agent is extensively used to treat various tumours. However, Cy at high doses would also lead to the impairment of mucosal immunity and gut microbiota in the intestine (Ahlmann & Hempel, 2016). In this study, we focused on the intestine and found that CSP ameliorated Cy-induced intestinal mucosal immunosuppression and microbial dysbiosis in mice.

$CD4^+$ T cells are critical for intestinal mucosal immunity. With the regulation of cytokines, naive $CD4^+$ T cells differentiate into different

Table 2
OTUs and α-diversity indices of gut microbiota in each group.

Groups	OTUs	Simpson	Chao1	ACE	Shannon
NC	1389.5 ± 207.2	0.9742 ± 0.007	1661.56 ± 294.83	1669.27 ± 271.45	7.79 ± 0.33
M	1472.67 ± 185.14	0.978 ± 0.02	1550.9 ± 283.6	1581.35 ± 314.35	8.13 ± 0.34
CSP	1723.83 ± 157.31*	0.9847 ± 0.004	2016.85 ± 201.4**	2046.8 ± 209.61**	8.58 ± 0.29*

Data are expressed as the mean ± SD (n = 6). $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with NC group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with M group.

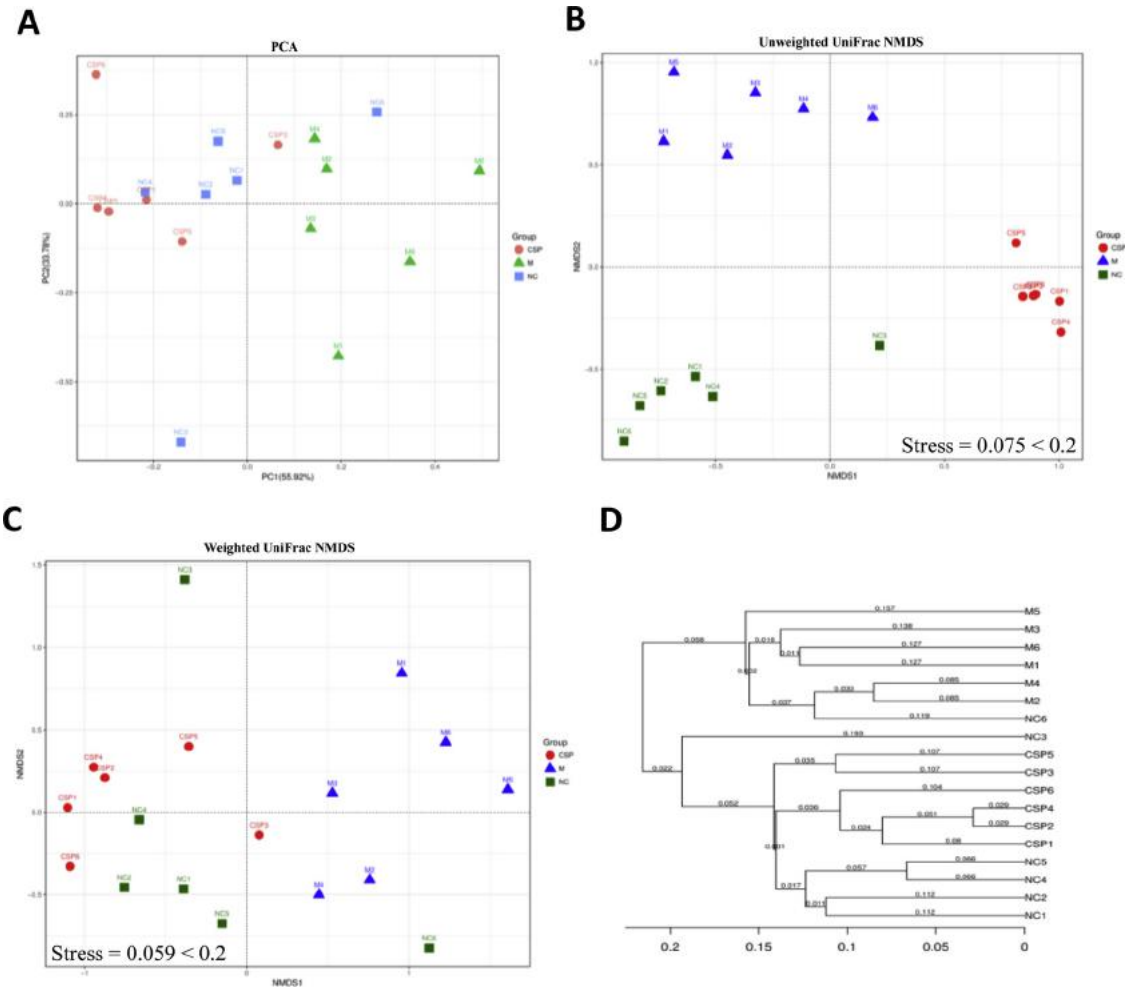

Fig. 5. β-diversity analysis of gut microbiota (n = 6 per group). (A) Principal component analysis (PCA). PC1 and PC2 represented two most principal factors affecting bacterial profiles and their probability (%) was showed on the axes. (B) Nonmetric multidimensional scaling (NMDS) analysis based on unweighted UniFrac distances. (C) NMDS analysis based on weighted UniFrac distances. Each point represented a sample and a shorter distance between the points indicated more similar microbial community structure of the samples. (D) Hierarchical clustering analysis based on weighted UniFrac distances. The number of branches represented the phylogenetic distance; the shorter the branches, the closer the evolutionary relationship.

Th subsets which is also associated with upregulation of master transcription factors (Zhu & Paul, 2008). IL-12 induces T-bet expression and Th1 cells differentiation which secretes IFN-γ and TNF-α (Wong, Lai, & Cheung, 2011). Th2 differentiation is a result of GATA-3 expression induced by IL-4 and IL-2, leading to the production of IL-4, IL-5 and IL-13 (Vahedi et al., 2013). TGF-β, IL-6 and IL-23 favors RORγt expression, which results in Th17 differentiation and make IL-17, IL-21, IL-22 (Korn et al., 2009). Treg differentiation and Foxp3 expression is induced by TGF-β, leading to the secretion of TGF-β and IL-10 (Workman, Szymczak-Workman, Collison, Pillai, & Vignali, 2009). Previous experiments have shown that *Ganoderma atrum* polysaccharides promoted IL-17A, IL-6, IL-10, TGF-β1, RORγt, Foxp3 levels in Cy-treated mice spleen tissues (Xiang et al., 2018) and natural *Cordyceps sinensis* polysaccharides elevated IL-4, IFN-γ, TNF-α, T-bet, GATA-3 mRNA levels in Cy-induced mice intestine (Chen, Wang, Fang, Dong, & Nie, 2019). However, compared with them, CSP could regulate more cytokines and transcription factors. Our findings showed that the production of IL-12, IFN-γ, IL-4, IL-13, IL-6, IL-17, IL-10, TGF-β3, TNF-α, IL-2, IL-21 and the mRNA levels of TNF-α, IL-2, IL-21 all significantly decreased in Cy-induced group compared to normal group as well as the expression of T-bet, GATA-3, RORγt and Foxp3, but all of them were upregulated after the treatment of CSP. These results demonstrated that CSP could alter mucosal immunosuppression induced by Cy and restore intestinal mucosal immunity by regulating Th1, Th2, Th17 and Treg cells differentiation via stimulating cytokines secretion and master transcription factors production.

Toll-like receptors (TLRs) are a type of pattern recognition receptors that play critical roles in the innate immune system, and exist in different cell types including epithelial cells and immune cells (Zhang et al., 2019). Together with secretory proteins MD-2 and CD14, they stimulate downstream signaling cascades which result in activation of NF-κB and induction of cytokine secretion (Kordjazy et al., 2018). NF-κB is a ubiquitous transcription factor and roles in the regulation of gene expression related to many cell processes. It can regulate the expression of cytokines and serves as a key responder to environment changes (Hayden & Ghosh, 2004). When NF-κB is activated, the inhibitory NF-κB protein IκBs will be phosphorylated by IKK, thus transferring the activated NF-κB to the nucleus (Zong et al., 2016). Therefore, we proposed the hypothesis that TLRs and NF-κB pathway were related to the regulation of transcription factors and cytokines. Our data indicated that CSP restored the mRNA levels of TLR-2, TLR-4, TLR-6, p-IκB-α expression in cytoplasm, and NF-κB p65 expression in nucleus, which were all inhibited by Cy in M group. The above results demonstrated that CSP stimulated cytokines secretion through TLRs and NF-κB pathway and thus affected Th cells differentiation.

Gut microbiota is closely related to intestinal mucosal immunity. Based on the ability of CSP to recover Cy-induced intestinal mucosal immunosuppression, we were inspired to investigate whether CSP could

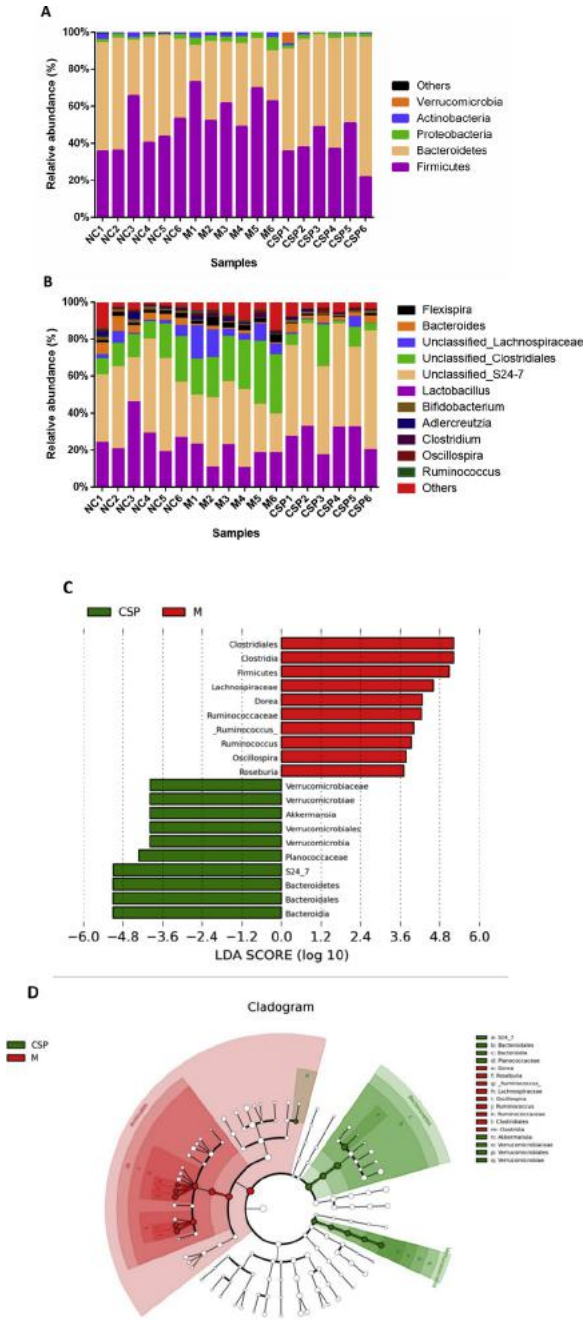

Fig. 6. CSP modulated microbial community composition and key phylotypes of gut microbiota in different groups (n = 6 per group). (A) Relative abundance of microbiota at the phylum level. (B) Relative abundance of microbiota at the genus level. (C) LDA score. Enriched taxa with an LDA score > 3.6 were shown in the histogram. The greater the LDA score was, the more significant the phylotype microbiota was in the comparison. (D) LEfSe taxonomic cladogram. The colored nodes from inner circle to outer circle represented the hierarchical relationship of all taxa from the phylum to the genus level. Taxa enriched in CSP group were showed in green while taxa enriched in M group were shown in red and taxa with nonsignifcant changes were colored in white. The diameter of each small circle represented the taxa abundance.

regulate gut microbiota. To explore this issue, SCFAs, which are important bridge between gut microbiota and intestinal mucosal immunity, were further investigated. SCFAs as the main end products of intestinal microbial fermentation of complex carbohydrates, play effective roles in regulating immune responses, maintaining epithelial barrier function, as well as inhibiting tumors (Maslowski & Mackay, 2011). They mainly contain acetic acid, propionic acid, butyric acid and valeric acid. Acetic acid can enhance reactive oxygen species production and phagocytosis, induce apoptosis and modulate the recruitment of neutrophils (Wang et al., 2019). Butyric acid is the major energy source for intestinal epithelial cells with the ability of regulating growth and immune response of epithelial cells and protecting intestinal related diseases (Han et al., 2018). As shown in our results, SCFAs all decreased significantly in M group compared with NC group and upregulated after CSP administration, especially in HD group. In addition, acetic acid and butyric acid were the top two kinds of SCFAs elevated by CSP, whereas the dietary fucoidan of *Acaudina molpadioides* showed a better effect on propionic acid (Shi et al., 2017). These data indicated that CSP could increase the levels of SCFAs, suggesting that gut microbiota in Cy-treated mice colon were influenced by CSP treatment.

Gut microbiota are composed of trillions of bacteria, which participate in numerous interactions with the host and play important roles in modulating host's immune system through bacterial components and metabolites (Ramakrishna, 2013). Recent years, varieties of polysaccharides have been proved to have the ability of regulating intestinal microbial ecology (Nie et al., 2019; Shang et al., 2018). Therefore, we investigated the effect of CSP on the Cy-treated mice gut microbiota by 16S rRNA sequencing technology. Based on the result of SCFAs, HD group was chose for further experiment and renamed as CSP group. α-diversity analysis demonstrated that CSP could enhance microbial community diversity, both microbial community richness and uniformity. Chao, ACE and Shannon indices increased in CSP group while M group had no obvious changes compared with NC group. Moreover, the data of OTUs also showed an increasing trend of microbial community diversity after CSP administration. β-diversity analysis, including PCA, NMDS analysis and Hierarchical clustering analysis revealed that the overall microbial community structure significantly changed after Cy treatment and CSP could attenuate these negative effects, the structure of CSP group was more similar to NC group.

Subsequently, more detailed taxonomic analysis of microbial community composition (phylum and genus level) was delivered. At the phylum level, *Firmicutes* and *Bacteroidetes* accounted for more than 90 % of the microbiota in all three groups. *Firmicutes* and *Bacteroidetes* can treat and scavenge dietary polysaccharides on the basis of carbohydrate-active enzymes, between which, *Bacteroidetes* are the primary saccharolytic bacteria in complex carbohydrates fermentation while *Firmicutes* prefer oligosaccharides (Cockburn & Koropatkin, 2016). In this study, *Firmicutes* increased and *Bacteroidetes* decreased in M group, and CSP reversed this trend which was compatible with the result that high-dose CSP promoted SCFAs production. This finding was similar to some other polysaccharides repored before (Shi et al., 2017; Tang et al., 2018). *Proteobacteria* are a major phylum of Gram-negative bacteria with numerous pathogenic microorganisms, such as *Helicobacter* and *Salmonella* (Baumler & Sperandio, 2016). Our results showed that CSP reduced the the abundance of *Proteobacteria* whereas Cy enhanced them. At the genus level, *Bifidobacterium* and *Lactobacillus* are the most common benchmark for the effectiveness of probiotics, with various beneficial effects for the host while *Clostridium* are usually harmful to the host's health (Khodaei, Fernandez, Fliss, & Karboune, 2016). *Flexispira* are closely linked to *Helicobacter*, which are pathogenic to the body (Cuccherini et al., 2000). *Bacteroides* can ferment polysaccharides to produce volatile fatty acids, which are reabsorbed through the large intestine and utilized as energy sources (Wexler, 2007). Different polysaccharides have different effects on gut microbiota. Squid ink polysaccharides decreased the abundance of *Ruminococcus*, *Bilophila*, *Oscillospira*, *Dorea*, *Mucispirillum* in Cy-treated mice (Lu et al., 2016). In contrast, CSP reduced the levels of *Clostridium*, *Flexispira* and recovered *Lactobacillus*, *Bifidobacterium* as well as *Bacteroides*. Moreover, LEfSe analysis was performed to identified the key phylotypes of gut microbiota among NC group, M group and CSP group. Results indicated that the major microbiota in CSP group were *Bacteroidetes*, *Bacteroidia*, *Bacteroidales* totally different from *Firmicutes*, *Clostridia*, *Clostridiales* in

M group. Overall, these above findings confirmed the potential of CSP as a prebiotics to restore gut microbial dysbiosis and regulate intestinal microbial ecology by improving microbial community diversity, modulating microbial community structure and composition in Cy-treated mice.

5. Conclusion

In conclusion, the present study suggested that CSP could recover Cy-induced intestinal mucosal immunosuppression by regulating Th cells differentiation via stimulating cytokines secretion and transcription factors production, which maybe associate with TLRs and NF-κB pathway. Moreover, CSP reversed the levels of SCFAs in colon and restored Cy-induced gut microbial dysbiosis by improving microbial community diversity, modulating microbial community structure and composition. These findings demonstrated the potential of CSP as a prebiotics to reduce side effects of Cy on intestinal mucosal immunity and gut microbiota, and provided theoretical foundation for the development of functional food based on cultured *C. sinensis*. However, more detailed mechanisms of CSP on ameliorating intestinal mucosal immunity and gut microbiota remain to be further explored.

Author contributions

Qiang Yu established the study. Shaoping Nie and Mingyong Xie designed the experiments. Mengxi Ying performed the experiments, analyzed the data and edited the manuscript. Bing Zheng and Hui Wang contributed to literature searches, article selections and western blot analysis. Junqiao Wang and Shuping Chen prepared the polysaccharides. All authors have read and approved the final manuscript.

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Declaration of Competing Interest

The authors declare no competing financial interest.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.carbpol.2020.115957.

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